```
• <strong>Digit <script type=”text/javascript”> document.location=’http://www.thinkdigit.com/’; </script></strong>
```

When the browser reads that, it will redirect to Digit's website. That is one of the most simple examples of code injection wherein the input received by the program actually tries to produce results that were unintended. In the web domain, this example would actually be called an XSS (Cross Site Scripting) attack.

An application working on the computer would receive and process inputs given to it by the user from time to time. In a fashion similar to that of the webpage, if the user tries to send an input to the program which would actually cause the program to do something different than it intended to, then it would be called 'code injection'. It's termed as code injection because the input given to the target program is usually a carefully crafted piece of executable code. Remember however, that a code injection for an application program is much more complicated and difficult than that for a web page due to the arcane format in which executable codes are expressed.

Address Space Layout Randomization is a method to prevent hackers from executing an injected code. Typically when a library (a DLL file) loads into the memory, it prefers to be put at a certain address in the memory. This is true for the executable files as well. Only if the location is unavailable, will Windows put it elsewhere. ASLR changes that. It randomizes the memory location where libraries are to be loaded every time Windows boots. This takes away one of the biggest advantages in the hands of hackers – predictability of memory address of a library. ASLR was introduced in Windows Vista.

New security features in Windows 8

While to an average user, these minute changes make an insignificant or zero difference, they do add up in the end to make Windows 8 more secure than its elder siblings. Let's talk now about some unnoticeable, yet significant security improvements in Windows 8:

Forced ASLR

Forced ASLR, a feature introduced in Windows 8, tries to force ASLR on every library that is to be loaded. However if the library doesn't have the information needed to leverage the ASLR function then Forced ASLR